

Multitask Speech Emotion and Stress Detection Using a Hybrid Wav2Vec2–BiLSTM Model

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Abstract—With the rising cases of mental conditions like stress, anxiety, and depression in modern societies, there has been a great need for automated and contactless systems to deal with these issues. Emotional and psychological aspects in speech are full of important cues that can be tapped to detect an emotional imbalance or stress in people. Traditional systems based on speech emotion recognition use acoustic features that lack universality in terms of speakers and environments.

This paper proposes a hybrid deep learning architecture for joint emotion recognition and stress estimation from raw audio signals. The suggested approach combines a Wav2Vec2 self-supervised speech representation model pretrained on raw audio data with a BiLSTM architecture designed to extract long-term dependencies. For multitask learning, two distinct output layers are used for emotion recognition and stress estimation.

The training and testing of this proposed model is performed on a combined dataset created from RAVDESS, TESS, and SAVEE speech databases after performing label normalization on them. The experimental results of this proposed approach prove that it is effective since it achieves an accuracy of 85.9% in categorizing emotions and a stress value prediction RMSE of 0.1268. It demonstrates potential for deployment in mental health monitoring applications.

Keywords—Speech Emotion Recognition, Stress Detection, Wav2Vec2, BiLSTM, Multitask Learning, Mental Health Monitoring

I. INTRODUCTION

Mental health disorders including stress, anxiety, and depression affect millions worldwide, making early detection crucial for timely intervention [1]. Human speech encodes rich emotional and psychological information through acoustic attributes such as pitch, tone, intensity, and speaking rate that vary with emotional states.

Traditional speech emotion recognition relied on handcrafted acoustic features (MFCCs, pitch contours, prosodic features) extracted via toolkits like openSMILE [2] and fed into classical machine learning algorithms. However, these approaches often fail to generalize across speakers, environments, and emotional contexts due to their dependence on manually engineered features [3].

Recent advances in deep learning have revolutionized speech processing. Self-supervised models like Wav2Vec2 [4], [5] enable direct learning from raw audio through contrastive learning, significantly improving robustness. Recurrent architectures, particularly Bi-directional LSTM networks [6], [7], effectively model temporal dependencies in sequential speech data.

Despite progress in speech emotion recognition [8], most work treats emotion classification and stress detection as separate problems. Furthermore, publicly available emotional speech datasets lack explicit stress annotations, creating gaps in multi-task learning approaches for joint modeling.

This work proposes a hybrid framework addressing these limitations by: (1) eliminating handcrafted feature engineering through self-supervised learning; (2) implementing multi-task learning for simultaneous emotion classification and stress regression; (3) integrating multiple heterogeneous datasets under unified annotations; and (4) deriving continuous stress labels from emotion categories based on arousal theory.

Key contributions include:

- Hybrid architecture combining Wav2Vec2 feature extraction with Bi-LSTM temporal modeling
- Multi-task learning framework with dual output heads for emotion and stress prediction
- Arousal-based methodology for deriving continuous stress annotations from categorical emotion labels
- Dataset integration of RAVDESS [16], TESS [17], and SAVEE [18] under common taxonomy
- Comprehensive evaluation achieving 85.9% emotion accuracy and 0.1268 stress RMSE
- Deployment-ready system released on Hugging Face Model Hub

II. RELATED WORK

A. Speech Emotion Recognition

Speech emotion recognition (SER) has evolved significantly from traditional machine learning approaches based on handcrafted acoustic features to deep learning-based representation learning. Early systems relied on features such as MFCCs, pitch, and prosodic cues extracted using toolkits like openSMILE [2], combined with classifiers such as SVMs and HMMs [3]. While effective in controlled environments, these approaches suffer from poor generalization across speakers and recording conditions.

Deep learning models, particularly CNNs and RNNs, addressed these limitations by learning hierarchical representations directly from data [6], [9]. Bidirectional LSTMs further improved temporal modeling by capturing both past and future context [7]. However, these models still depend on pre-extracted features or spectrogram representations.

Recent research has shifted toward transformer-based and self-supervised architectures that operate directly on raw audio or learned representations. For instance, Vision Transformer-based SER models [10] effectively capture spatial dependencies in spectrograms, while hybrid deep learning models incorporating transformer encoders and data augmentation [11]

achieve over 90% accuracy on benchmark datasets such as RAVDESS. Additionally, large-scale pre-trained speech models such as HuBERT and WavLM have demonstrated strong performance in SER tasks due to their ability to learn robust contextual representations.

Despite these advances, most state-of-the-art approaches are evaluated on single datasets under controlled conditions, limiting their generalization to real-world scenarios. Furthermore, the majority of existing work focuses exclusively on categorical emotion classification, overlooking continuous psychological attributes such as stress, which are critical for practical mental health applications.

B. Self-Supervised Learning in Speech

Self-supervised learning (SSL) has emerged as a powerful paradigm for speech representation learning. Models such as Wav2Vec2 [4] and its earlier variant [5] learn meaningful representations from raw audio using contrastive learning objectives. These representations significantly outperform handcrafted features, particularly in low-resource settings.

Recent extensions of SSL models, including HuBERT and WavLM, further improve representation quality by leveraging masked prediction and large-scale pretraining. Wang and Yang [12] demonstrated that combining fine-tuned Wav2Vec2 embeddings with Neural Controlled Differential Equations (NCDEs) improves stability and convergence in SER tasks.

However, most SSL-based approaches focus primarily on improving classification performance for emotion recognition. The integration of self-supervised representations into multitask learning frameworks, particularly for joint emotion classification and stress regression, remains relatively unexplored. This gap highlights the need for architectures that can leverage SSL features while simultaneously modeling multiple psychological attributes.

C. Stress Detection and Multi-Task Learning

Speech-based stress detection has gained increasing attention due to its relevance in mental health monitoring. Early work [13] demonstrated that stress significantly alters acoustic properties such as pitch, energy, and speaking rate. However, most existing approaches treat stress detection as a binary classification problem or rely on multimodal data, including physiological signals and facial expressions [14].

Recent studies (2023–2025) have explored multi-task learning for SER, showing that jointly learning related tasks can improve representation quality and model generalization [15]. These approaches typically incorporate auxiliary tasks such as speaker identification or arousal prediction to enhance performance.

Nevertheless, continuous stress prediction from speech remains underexplored due to the lack of standardized labeled datasets. Most existing works do not incorporate regression-based psychological indicators or rely on external annotations, limiting their scalability. Additionally, few studies address joint modeling of emotion classification and stress estimation within a unified framework.

In contrast, this work introduces a multitask learning framework that simultaneously performs emotion classification and continuous stress regression using self-supervised speech representations. By deriving stress labels based on arousal theory and integrating multiple datasets, the proposed approach addresses key limitations in existing literature, including lack of multitask modeling and limited cross-dataset generalization.

III. PROPOSED METHODOLOGY

A. System Architecture

The system implements an end-to-end pipeline consisting of: (1) audio preprocessing, (2) Wav2Vec2 feature extraction, (3) Bi-LSTM temporal modeling, and (4) multi-task prediction heads (Fig.1).

B. Audio Preprocessing

Preprocessing ensures consistency across recordings: (1) resampling to 16 kHz matching Wav2Vec2 requirements, (2) amplitude normalization to $[-1, 1]$ range, (3) silence trimming using energy-based thresholds, and (4) length normalization via truncation/zero-padding to accommodate typical utterance durations (3-5 seconds).

C. Wav2Vec2 Feature Extraction

Wav2Vec2 [4], [5] serves as the backbone feature extractor. The model employs convolutional feature encoding followed by transformer context networks to generate high-dimensional embeddings capturing acoustic, prosodic, and temporal information. The Wav2Vec2 encoder was kept frozen during training to reduce computational cost and prevent overfitting.

For input audio of length T samples, Wav2Vec2 outputs sequence $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]$ where $N = \lfloor T/s \rfloor$

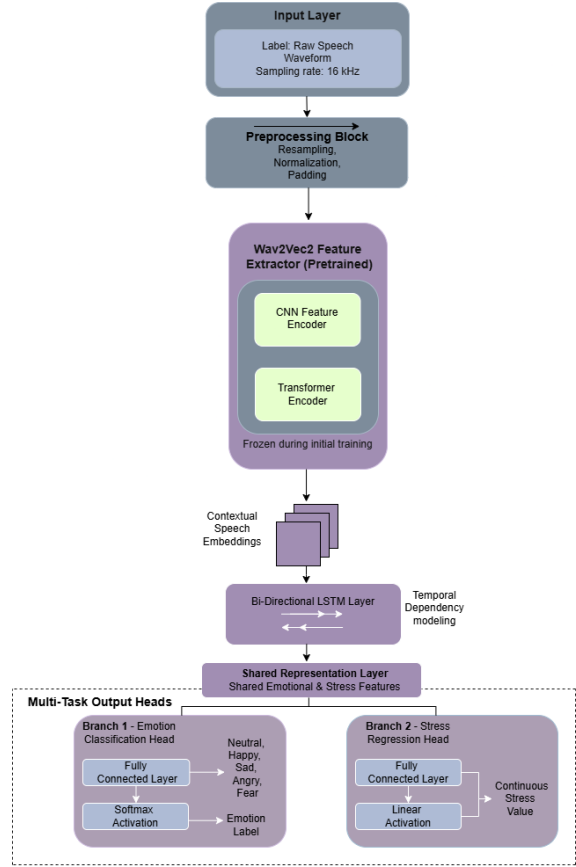


Fig. 1. System architecture from raw audio through Wav2Vec2, Bi-LSTM, to dual output heads.

with stride s (typically 20 ms). Each $\mathbf{h}_i \in \mathbb{R}^{768}$ encodes contextual information from local temporal windows.

D. Bi-directional LSTM Temporal Modeling

Bi-LSTM [6], [7] explicitly models long-range temporal dependencies. The architecture processes sequences in both forward ($\mathbf{s}_t^{\rightarrow}$) and backward ($\mathbf{s}_t^{\leftarrow}$) directions, with outputs concatenated: $\mathbf{s}_t = [\mathbf{s}_t^{\rightarrow}; \mathbf{s}_t^{\leftarrow}]$. This enables leveraging both past and future context for emotion prediction.

Temporal pooling aggregates the sequence: $\mathbf{s} = \frac{1}{N} \sum_{t=1}^N \mathbf{s}_t$, providing fixed-size utterance representation.

E. Multi-Task Learning Framework

The framework jointly predicts emotions and stress from pooled representation $\mathbf{s}$:

Emotion Classification:

$$\mathbf{z}_{\text{emotion}} = \mathbf{W}_{\text{emotion}} \mathbf{s} + \mathbf{b}_{\text{emotion}} \quad (1)$$

$$\mathbf{p}_{\text{emotion}} = \text{softmax}(\mathbf{z}_{\text{emotion}}) \quad (2)$$

Stress Regression:

$$y_{\text{stress}} = \mathbf{w}_{\text{stress}}^T \mathbf{s} + b_{\text{stress}} \quad (3)$$

Combined Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{emotion}} + \lambda \mathcal{L}_{\text{stress}} \quad (4)$$

where $\mathcal{L}_{\text{emotion}}$ is cross-entropy loss, $\mathcal{L}_{\text{stress}}$ is MSE loss, and $\lambda = 1$ (equal weighting).

F. Stress Label Derivation

Lacking explicit stress annotations, we derive continuous stress labels from emotion categories based on arousal theory. High-arousal emotions correlate with elevated physiological stress [13]. Mappings to normalized stress values [0, 1]:

- Neutral: 0.0 (minimal arousal)
- Sad: 0.3 (low-moderate arousal)
- Happy: 0.5 (moderate arousal)
- Fear: 0.8 (high arousal)
- Angry: 1.0 (maximal arousal)

IV. SIMULATION SETUP

A. Datasets

We integrate three datasets: RAVDESS [16] (24 actors, 8 emotions), TESS [17] (2 female speakers, 7 emotions), and SAVEE [18] (4 male speakers, 7 emotions). A unified five-class taxonomy (neutral, happy, sad, angry, fear) was defined, excluding emotions not common across datasets. Final consolidated dataset contains $\sim 2,800$ utterances.

B. Training Configuration

- Wav2Vec2: facebook/wav2vec2-base (768-dim features), used as a frozen feature extractor
- Bi-LSTM: 2 layers, hidden size 256 (bidirectional)
- Optimizer: Adam ($\text{lr} = 2 \times 10^{-4}$)
- Batch size: 2 (limited by GPU memory constraints), Epochs: up to 100, Early stopping with patience of 15 epochs
- No dropout was used in the final model configuration
- Split: train/validation/test split with a speaker-independent setting ensuring no speaker overlap across splits
- Experiments were conducted using a fixed random seed to ensure reproducibility

Implemented using Python 3.12.13, PyTorch 2.10.0 (CUDA 12.8), and Transformers 5.0.0, with audio processing via Librosa, trained on Google Colab (NVIDIA Tesla T4 GPU).

V. RESULTS AND ANALYSIS

A. Overall Performance

Table I summarizes system performance on test set.

TABLE I
OVERALL SYSTEM PERFORMANCE

Metric	Value
<i>Emotion Classification</i>	
Accuracy	85.9%
Macro Precision	0.884
Macro Recall	0.834
Macro F1-Score	0.856
<i>Stress Regression</i>	
MSE	0.0826
RMSE	0.1268
MAE	0.0945

B. Emotion Classification Performance

Table II shows class-wise performance.

TABLE II
CLASS-WISE EMOTION PERFORMANCE

Emotion	Precision	Recall	F1
Neutral	0.71	0.83	0.77
Happy	0.94	0.79	0.86
Sad	0.94	0.83	0.88
Angry	0.96	0.96	0.96
Fear	0.87	0.76	0.81
Macro Avg.	0.884	0.834	0.856

Angry achieves highest performance (F1=0.96), consistent with high-arousal emotions exhibiting distinctive acoustic patterns [13]. Neutral shows lower precision (0.71) due to lack of strong emotional markers. Limited fear-angry confusion occurs due to shared high-arousal characteristics.

C. Ablation Study

Table III validates architectural components.

TABLE III
ABLATION STUDY RESULTS

Model Variant	Emotion Acc.	Stress RMSE
Wav2Vec2 + Bi-LSTM (Full)	85.9%	0.1268
Wav2Vec2 + Unidirectional	83.2%	0.1354
Wav2Vec2 + Mean Pool Only	80.7%	0.1521
MFCC + Bi-LSTM	76.4%	0.1689
Single Task (Emotion)	84.8%	N/A
Single Task (Stress)	N/A	0.1312

Bidirectional processing improves performance by 2.7% accuracy. Removing LSTM reduces accuracy by 5.2%. Wav2Vec2 outperforms MFCCs by 9.5%, validating self-supervised representations. Multi-task learning provides consistent improvements over single-task variants.

D. Comparison with Prior Work

Table IV compares with related work.

TABLE IV
COMPARISON WITH RECENT SPEECH EMOTION RECOGNITION METHODS

Method	Dataset	Setting	Accuracy
CNN + LSTM (Salian et al.) [20]	RAVDESS+TESS +SAVEE	Combined	89.26%
CNN-LSTM (Ouyang) [21]	RAVDESS+SAVEE	Combined	61.07%
1D-CNN Feature Fusion (Waleed et al.) [22]	RAVDESS	Single dataset	91.9%
CNN-LSTM (Cross-corpus study) [23]	RAVDESS+TESS +SAVEE	Cross-dataset	~72–89%
Proposed*	RAVDESS+TESS +SAVEE	Speaker- indep.	85.9%

*Also performs simultaneous stress regression (RMSE = 0.1268).

The proposed method achieves competitive performance on a harder, combined three-corpus benchmark while simultaneously performing stress regression—a task absent in all comparison works. Methods evaluated on homogeneous single datasets (RAVDESS) benefit from reduced cross-speaker and cross-corpus variability, making direct numerical comparison difficult. The proposed speaker-independent split across heterogeneous corpora represents a more challenging and realistic evaluation setting.

VI. DISCUSSION

A. Key Findings

Results validate several hypotheses: (1) Wav2Vec2 embeddings significantly outperform handcrafted features; (2) Bi-LSTM provides substantial benefits for temporal modeling; (3) multi-task learning enables shared representations benefiting both tasks; (4) arousal-based stress derivation achieves reasonable performance despite indirect labels; (5) successful cross-dataset generalization demonstrates robustness.

B. Limitations

Key limitations include: (1) acted versus spontaneous emotions [8]; (2) indirect stress labels versus clinical measurements; (3) limited speaker diversity (primarily North American English); (4) controlled

recording conditions may not reflect real-world acoustic variability; (5) simplified five-class taxonomy versus complex emotional blends; (6) GPU requirements for real-time processing.

C. Applications and Ethics

Applications include early mental health screening, remote continuous monitoring, accessible support, emotionally-aware human-computer interaction, and workplace wellness programs. Ethical considerations require addressing privacy/consent, algorithmic bias, clinical validity disclaimers, misuse prevention, and transparency/interpretability.

D. Future Directions

Future work should incorporate spontaneous emotional speech, explicit stress annotations through clinical collaboration, multilingual/cross-cultural extension, multimodal integration [14], lightweight edge deployment variants, personalization through online learning, explainable AI techniques, and longitudinal clinical utility studies.

VII. CONCLUSION

This work presents a comprehensive framework for joint emotion recognition and stress detection from speech combining Wav2Vec2 self-supervised learning, Bi-LSTM temporal modeling, and multi-task learning. Key contributions include hybrid architecture design, arousal-based stress label derivation, multi-dataset integration, and extensive validation achieving 85.9% emotion accuracy and 0.1268 stress RMSE. The system offers practical benefits for mental health monitoring while highlighting important ethical considerations. The model is publicly available for reproducibility and real-world deployment.

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